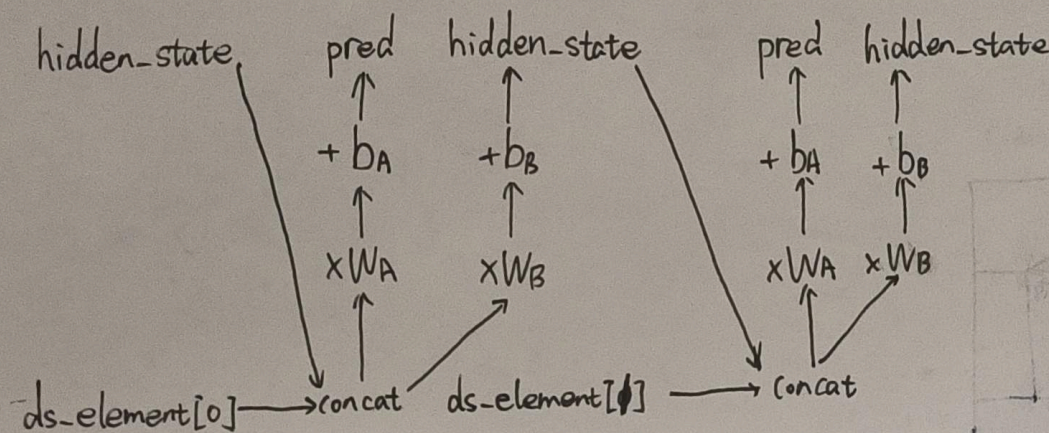


ex04

Gradient of Negative Conditional Log-Likelihood

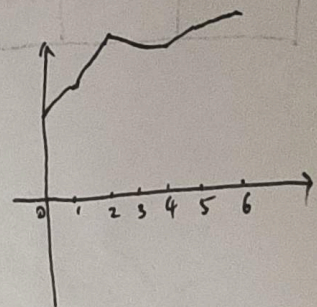
$$\begin{aligned}\nabla_w \mathcal{L}(w) &= - \sum_{n=1}^N [\nabla_w \psi(x^n, y^n, w) - \frac{\sum_y y \exp\{\psi(x^n, y, w)\} \nabla_w \psi(x^n, y, w)}{\sum_y \exp\{\psi(x^n, y, w)\}}] \\ &= - \sum_{n=1}^N [\nabla_w \psi(x^n, y^n, w) - \sum_y \frac{\exp\{\psi(x^n, y, w)\}}{\sum_y \exp\{\psi(x^n, y, w)\}} \nabla_w \psi(x^n, y, w)] \\ &= - \sum_{n=1}^N [\nabla_w \psi(x^n, y^n, w) - \sum_y p(y|x^n, w) \nabla_w \psi(x^n, y, w)]\end{aligned}$$

Inference Unrolling



Surface Integration

x_i	0	1	2	3	4	5	6
$\frac{dz}{dx}(x_i)$	1.5	2	-1	0	1	1	0
$z(x_i)$	7.5	9	11	10	10	11	12



Volumetric Fusion Formulation

$$WSE = \sum_i w_i (d_i - D)^2$$

$$\frac{\partial WSE}{\partial D} = \sum_i 2w_i (d_i - D^*) = 0$$

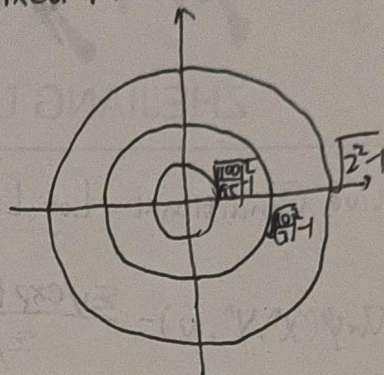
$$\sum_i -w_i d_i + \sum_i w_i D = 0$$

$$D = \frac{\sum_i w_i d_i}{\sum_i w_i}$$

$$b) D_{i+1}(x) = \frac{\sum_{n=1}^{i+1} w_n(x) d_n(x)}{W_i(x) + w_{i+1}(x)} = \frac{W_{i+1}(x) d_{i+1}(x) + D_i(x) W_i(x)}{W_i(x) + w_{i+1}(x)}$$

Drawing the p/q Reflectance Map for a Fixed R

$$a) R(p, q) = \frac{ps_x + qs_y + s_z}{\sqrt{p^2 + q^2 + 1}} = \frac{1}{\sqrt{p^2 + q^2 + 1}}$$



$$b) \frac{g}{nq} = \frac{z}{1+nz}$$

$$g = \frac{znq}{1+nz} = \frac{zn}{\frac{1}{q} + n} = \frac{zn}{\frac{1}{\sqrt{p^2 + q^2 + 1}} + n} = \frac{zn}{\frac{1 + n\sqrt{p^2 + q^2 + 1}}{\sqrt{p^2 + q^2 + 1}}} = \frac{zn\sqrt{p^2 + q^2 + 1}}{1 + n\sqrt{p^2 + q^2 + 1}}$$

$$f = \frac{z}{np} = \frac{z}{1+nz}$$

$$f = \frac{zn}{1+nz} = \frac{zn}{\frac{1}{p} + n} = \frac{zn}{\frac{1}{\sqrt{p^2 + q^2 + 1}} + n} = \frac{zn\sqrt{p^2 + q^2 + 1}}{1 + n\sqrt{p^2 + q^2 + 1}}$$

$$c) f^2 + g^2 = \frac{4(p^2 + q^2 + 1)}{(\sqrt{p^2 + q^2 + 1} + 1)^2} = \frac{4R^2}{(R+1)^2} = \frac{4}{(1+R)^2}$$

Marching Cubes

a)

